

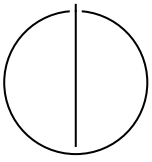
SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
CHAIR OF HEALTH INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Lukas Alex Jochim





SCHOOL OF COMPUTATION, INFORMATION AND TECHNOLOGY — CHAIR OF HEALTH INFORMATICS

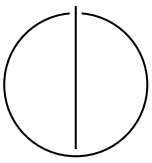
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Decoding Music Perception: EEG-Based Genre Classification: Knowledge Base

Dekodierung der Musikwahrnehmung: EEG-basierte Genre-Klassifizierung

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Submission Date:	01.09.2025



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, 01.09.2025 Lukas Alex Jochim

Acknowledgments

Abstract

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1 Introduction

1.1 Motivation

The human brain is not only one of the most complex systems within the human body but also one of the most important ones. The brain manages various functionalities like emotion, memory, attention or sensory processing just to name a few. Apart from that, the brain plays a significant role in how we process musical pieces. Independent of culture and age, music is a universal tool to infleunce one's emotions, concentration and cognition. Furthermore, the brain gives us the unique ability to retrieve many different features out of music like melody extraction or the identification of an artist or a specific genre. Depending on the style of music, the brain reacts differently. However, although the brain is central for cognitive processes such as listening to music, the core mechanisms actually remain unknown and currently are a black-box.

To enlighten this black box, medical advancements like functional magnetic resonance imaging (fMRI) or electroencephalogram (EEG) arrived, which are able to measure brain activity. Therefore, combining EEG and music have the ability to help understanding better on how our brain functions. By further adding means of Machine Learning (ML), a whole new interdisciplinary field opens up, bridging the gap between neurosciences and Computer science.

Regarding this, there is a growing research interest in the affective influence of music on the brain. However, classifying different music genres on the basis of EEG signals remains underrepresented. This is exactly where this thesis aims to close the gap, by investigating whether it is possible to classify various musical genres depending on the EEG signals acquired using state-of-the-art ML techniques.

The research results have high potential in the domain of medical computer science.

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